

SIMATS ENGINEERING

CO- 2: AT – 2: Simulation Task - Medium

COURSE NAME: Advance Wireless Communication Systems /
5G / 6G Communication Systems
COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 2 - *Analyse LTE and 5G wireless network architectures, including carrier aggregation, MIMO techniques, beamforming strategies, and service-based core network functional entities. (BL4)*

Assessment Tool Weightage: 35%

Each student should simulate one question and should upload the results in Github account along with the necessary outputs and codes. This assessment tool is evaluated for 50 marks.

QUESTIONS:	Total Marks: 50	Duration: 90 Mins
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1. Simulation of digital beamforming under user direction, beam width, SINR, antenna spacing, and channel condition (BL4)
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Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding Model Design	20%	Demonstrates strong understanding of underlying concepts in the simulation 5	Shows good understanding with minor gaps 4	Partial understanding; some misconceptions 3	Lacks understanding of key concepts 1-2
Tool Usage	20%	Uses simulation tools effectively with correct setup and execution 5	Uses tools with minor errors or guidance 4	Limited ability to use tools properly 3	Unable to use simulation tools 1-2
Analysis of Results	20%	Provides clear, logical analysis and meaningful interpretation of results 5	Analysis is reasonable with minor gaps 4	Superficial analysis; limited interpretation 3	Unable to analyze or interpret results 1-2
Documentation	20%	Presents results clearly with proper documentation 5	Documentation is adequate with minor issues 4	Poorly organized or incomplete documentation 3	No proper documentation or presentation 1-2

Criteria	Question Number :			
	Level 4	Level 3	Level 2	Level 1
Conceptual Understanding Model Design				
Tool Usage				
Analysis of Results				
Documentation				

clc;

```

clear;
close all;

%% Digital Beamforming Simulation
% Parameters:
% User direction, Beam width, SINR, antenna spacing, channel condition

%% 1. System Parameters
N = 8;           % Number of antenna elements
fc = 3.5e9;      % Carrier frequency = 3.5 GHz
c = 3e8;         % Speed of light
lambda = c/fc;   % Wavelength

d = lambda/2;    % Antenna spacing
SNR_dB = 20;     % Signal-to-noise ratio
INR_dB = 10;     % Interference-to-noise ratio

user_angle = 30; % Desired user direction (degrees)
interference_angle = -20; % Interference direction (degrees)

%% 2. Channel Condition
% Choose:
% 'LOS' = Line-of-Sight
% 'NLOS' = Non-Line-of-Sight
channel_condition = 'LOS';

%% 3. Antenna Array
n = (0:N-1).';

%% 4. Steering Vector Function
steering_vector = @(angle) ...
    exp(1j*2*pi*d/lambda*n*sind(angle));

%% 5. Desired User Channel
h_user = steering_vector(user_angle);

%% 6. Interference Channel
h_interference = steering_vector(interference_angle);

%% 7. Channel Condition
if strcmpi(channel_condition,'LOS')
    % LOS channel
    h_user = h_user;
    h_interference = h_interference;

```

```

elseif strcmpi(channel_condition,'NLOS')
    % NLOS channel with random multipath
    h_user = (randn(N,1) + 1j*randn(N,1))/sqrt(2);
    h_interference = (randn(N,1) + 1j*randn(N,1))/sqrt(2);
else
    error('Invalid channel condition');
end

%% 8. Beamforming Weight
% Maximum Ratio Transmission / Receive Beamforming
w = h_user / norm(h_user);

%% 9. Beam Pattern
angles = -90:0.1:90;

beam_response = zeros(size(angles));

for k = 1:length(angles)

    a = steering_vector(angles(k));

    beam_response(k) = abs(w' * a)^2;

end

% Normalize beam pattern
beam_response = beam_response ./ max(beam_response);

beam_dB = 10*log10(beam_response + 1e-12);

%% 10. Calculate Beam Width
% Find -3 dB points
idx = find(beam_dB >= -3);

beam_width = angles(idx(end)) - angles(idx(1));

fprintf('\n===== DIGITAL BEAMFORMING RESULTS =====\n');

fprintf('Number of antennas      : %d\n',N);
fprintf('Carrier frequency      : %.2f GHz\n',fc/1e9);
fprintf('Wavelength              : %.4f m\n',lambda);
fprintf('Antenna spacing         : %.4f m (%.2f lambda)\n',d,d/lambda);
fprintf('User direction          : %.2f degrees\n',user_angle);

```

```
fprintf('Interference direction : %.2f degrees\n',interference_angle);
fprintf('Channel condition      : %s\n',channel_condition);
fprintf('Approx. Beam width     : %.2f degrees\n',beam_width);
```

%% 11. Signal and Interference Powers

```
SNR = 10^(SNR_dB/10);
```

```
INR = 10^(INR_dB/10);
```

```
signal_power = abs(w' * h_user)^2;
```

```
interference_power = abs(w' * h_interference)^2;
```

```
noise_power = signal_power/SNR;
```

%% 12. SINR Calculation

```
SINR = signal_power / ...
```

```
(interference_power + noise_power);
```

```
SINR_dB = 10*log10(SINR);
```

```
fprintf('Signal Power          : %.4f\n',signal_power);
```

```
fprintf('Interference Power     : %.4f\n',interference_power);
```

```
fprintf('Noise Power              : %.4f\n',noise_power);
```

```
fprintf('SINR                     : %.2f dB\n',SINR_dB);
```

%% 13. Plot Beam Pattern

```
figure;
```

```
plot(angles,beam_dB,'LineWidth',2);
```

```
grid on;
```

```
xlabel('Angle (degrees)');
```

```
ylabel('Normalized Gain (dB)');
```

```
title('Digital Beamforming Radiation Pattern');
```

```
xline(user_angle,'--','User');
```

```
xline(interference_angle,'--','Interference');
```

```
ylim([-40 5]);
```

%% 14. Effect of Antenna Spacing

```
spacing_values = [0.25 0.5 0.75 1.0];
```

```
figure;
```

```
hold on;
```

```

for s = 1:length(spacing_values)

    d_temp = spacing_values(s)*lambda;

    response = zeros(size(angles));

    for k = 1:length(angles)

        a = exp(1j*2*pi*d_temp/lambda*n*sind(angles(k)));

        response(k) = abs(w' * a)^2;

    end

    response = response/max(response);

    response_dB = 10*log10(response + 1e-12);

    plot(angles,response_dB,'LineWidth',1.5);

end

grid on;

xlabel('Angle (degrees)');
ylabel('Gain (dB)');
title('Effect of Antenna Spacing');

legend('0.25\lambda','0.5\lambda','0.75\lambda','1.0\lambda');

ylim([-40 5]);

%% 15. Effect of User Direction
user_angles = -60:5:60;
SINR_values = zeros(size(user_angles));

for u = 1:length(user_angles)

    hu = steering_vector(user_angles(u));

    wu = hu/norm(hu);

    desired = abs(wu' * hu)^2;

```

```

interference = abs(wu' * h_interference)^2;

noise = desired/SNR;

SINR_values(u) = ...
    10*log10(desired/(interference + noise));

end

%% 16. Plot SINR vs User Direction
figure;

plot(user_angles,SINR_values,'o-','LineWidth',2);

grid on;

xlabel('User Direction (degrees)');
ylabel('SINR (dB)');
title('SINR vs User Direction');

%% 17. Channel Condition Comparison
channel_types = {'LOS','NLOS'};
SINR_channel = zeros(1,2);

for ch = 1:2

    if strcmp(channel_types{ch},'LOS')

        h = steering_vector(user_angle);
        hi = steering_vector(interference_angle);

    else

        h = (randn(N,1)+1j*randn(N,1))/sqrt(2);
        hi = (randn(N,1)+1j*randn(N,1))/sqrt(2);

    end

    wb = h/norm(h);

    Ps = abs(wb' * h)^2;
    Pi = abs(wb' * hi)^2;
    Pn = Ps/SNR;

```

```

        SINR_channel(ch) = ...
            10*log10(Ps/(Pi+Pn));

end

%% 18. Channel Condition Plot
figure;

bar(SINR_channel);

grid on;

set(gca,'XTickLabel',channel_types);

xlabel('Channel Condition');
ylabel('SINR (dB)');
title('SINR for Different Channel Conditions');

%% 19. Display Final Results
fprintf('\n===== CHANNEL COMPARISON =====\n');

fprintf('LOS SINR : %.2f dB\n',SINR_channel(1));
fprintf('NLOS SINR : %.2f dB\n',SINR_channel(2));

fprintf('\nSimulation completed successfully.\n');

```




